



Vectors · Mark Scheme

Core Pure 1 Chapter 9: Vectors. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Lines and the scalar product

- 1 Direction $\vec{AB} = \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$ M1; $\mathbf{r} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$ A1.
- 2 Scalar product $2 - 4 - 2 = -4$ M1; $\cos \theta = \frac{-4}{3\sqrt{18}}$, $\theta = 108.3^\circ$ M1; acute angle 71.7° A1.
- 3 $\frac{x-2}{3} = \frac{y+1}{5} = \frac{z-4}{-2}$ B1; substitute: $\frac{6}{3} = \frac{10}{5} = \frac{-4}{-2}$ M1; all equal 2, so the point lies on l A1.
- 4 $1 + \lambda = 2 + \mu$, $\lambda = 4 - 2\mu$, $2 - \lambda = -3 + 3\mu$ M1; solving two equations: $\lambda = 2$, $\mu = 1$ A1; check third: $2 - 2 = 0 = -3 + 3$, consistent, so the lines intersect M1; $(3, 2, 0)$ A1.
- 5(a) $1 + 2\lambda = 3 + \mu$, $2 - \lambda = \mu$, $-1 + \lambda = 2 - 2\mu$ M1; solve two equations M1; $\lambda = \frac{4}{3}$, $\mu = \frac{2}{3}$ A1; third: $-1 + \frac{4}{3} = \frac{1}{3}$ but $2 - \frac{4}{3} = \frac{2}{3}$, contradiction, so no intersection A1.
- 5(b) The direction vectors are not multiples of each other, so the lines are not parallel; not parallel and not intersecting means skew B1.
- 5(c) $\cos \theta = \frac{2-1-2}{\sqrt{6}\sqrt{6}}$ M1; $= -\frac{1}{6}$, $\theta = 99.6^\circ$ A1; acute angle 80.4° A1.

Section B – Planes and the vector product

- 6 $4x + y - 2z = 4(2) + (-1) - 2(3)$ M1; $4x + y - 2z = 1$ A1.
- 7 Correct method for the vector product M1; $\begin{pmatrix} 4 \\ -5 \\ -6 \end{pmatrix}$ A1; $4 - 10 + 6 = 0$ and $12 + 0 - 12 = 0$ A1.
- 8 $\mathbf{n} = \begin{pmatrix} 4 \\ -5 \\ -6 \end{pmatrix}$, $p = \mathbf{n} \cdot \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = 4 - 12$ M1; $\mathbf{r} \cdot \begin{pmatrix} 4 \\ -5 \\ -6 \end{pmatrix} = -8$ A1; $4x - 5y - 6z = -8$ A1.
- 9 $\vec{AB} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}$, $\vec{AC} = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$ B1; $\vec{AB} \times \vec{AC} = \begin{pmatrix} 10 \\ -2 \\ 4 \end{pmatrix}$ M1; so $\mathbf{n} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$ and $5x - y + 2z = 5 + 0 + 2$ M1; $5x - y + 2z = 7$ A1.
- 10(a) Vector product method M1; $\begin{pmatrix} -6 \\ -2 \\ 2 \end{pmatrix}$, or any multiple such as $\begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$ A1.
- 10(b) $\begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} = \vec{AP}$ with $A(2, 1, 0)$ on l , so this vector lies in Π B1; $3x + y - z = 3(2) + 1 - 0$ M1; $3x + y - z = 7$ A1.
- 10(c) $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}$ B1.
- 10(d) $(0, 0, -7)$: $0 + 0 + 7 = 7$, so it lies in Π B1; $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} = 3 - 2 - 1 = 0$, so m is parallel to Π , hence lies in it B1.



Section C – Intersections and angles with planes

11 $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = 0$, so the line is parallel to the plane $B1$; $(1, 0, 2)$ gives $1 - 0 + 2 = 3 \neq 7$, so it does not lie in the plane $B1$.

12 $(1 + 2t) + 2(2 - t) + 3t = 11$ $M1$; $5 + 3t = 11$, $t = 2$ $A1$; $(5, 0, 6)$ $A1$.

13 $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = 4$ $M1$; $\cos \alpha = \frac{4}{9}$ gives the angle with the normal, $\alpha = 63.6^\circ$ $A1$; angle with the plane $= 90^\circ - 63.6^\circ = 26.4^\circ$ $A1$.

14 Line through P along the normal: $\mathbf{r} = \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ $M1$; $(5 + t) + 2(3 + 2t) + 2(4 + 2t) = 1 \Rightarrow 19 + 9t = 1$ $M1$; $t = -2$, foot $(3, -1, 0)$ $A1$; reflection at $t = -4$: $(1, -5, -4)$ $A1$.

15(a) $\overrightarrow{AB} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ $M1$; $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + t \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ $A1$.

15(b) $(1 + 2t) + 4(2 + t) + 8(3 - 2t) = 13$ $M1$; $33 - 10t = 13$, $t = 2$ $A1$; $(5, 4, -1)$ $A1$.

15(c) $\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 4 \\ 8 \end{pmatrix} = -10$, with magnitudes 3 and 9 $M1$; $\sin \theta = \frac{10}{27}$ $M1$; $\theta = 21.7^\circ$ $A1$.

Section D – Distances

16 $\frac{|2 - 2 + 6 - 10|}{\sqrt{4 + 1 + 4}} M1$; $= \frac{4}{3}$ $A1$.

17 $|\mathbf{n}| = \sqrt{4 + 9 + 36} = 7$ $B1$; distance $= \frac{|4 - (-10)|}{7} M1$; $= 2$ $A1$.

18 $\frac{|2k + 1 - 4 - 1|}{3} = 3$ $M1$; $2k - 4 = \pm 9$ $A1$; $k = \frac{13}{2}$ or $k = -\frac{5}{2}$ $A1$.

19 $\mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ $M1$ $A1$; vector between points on the lines $\begin{pmatrix} -1 \\ 1 \\ 5 \end{pmatrix}$, distance $= \frac{\left| \begin{pmatrix} -1 \\ 1 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right|}{\sqrt{3}} M1$;
 $= \frac{3}{\sqrt{3}} = \sqrt{3}$ $A1$.

20(a) $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} M1$; $= \begin{pmatrix} 3 \\ -6 \\ -1 \end{pmatrix} A1$.

20(b) $\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} B1$; distance $= \frac{|6 + 18 - 5|}{\sqrt{46}} M1$; $= \frac{19}{\sqrt{46}} = 2.80$ m $A1$.

20(c) Normal $\begin{pmatrix} 3 \\ -6 \\ -1 \end{pmatrix}$ through $(1, 2, -1)$: $3 - 12 + 1 = -8$ $M1$; $3x - 6y - z = -8$ $A1$.

20(d) E.g. the pipe and cable have thickness, so the actual clearance is less; they may not be perfectly straight $B1$.



Roar-level challenge

21 $\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = -1 \neq 0$, so not parallel $B1$;

general point $(1 + 2t, t, 2 - 2t)$ $M1$; distance $\frac{|2(1 + 2t) - t + 2(2 - 2t) - 1|}{3} = \frac{|5 - t|}{3} = 3$ $M1$; $t = -4$ or $t = 14$ $A1$;
 $(-7, -4, 10)$ and $(29, 14, -26)$ $A1$.